

4. (25 marks) For $(a, b) \in \mathbb{R} \times \mathbb{R}_+$, consider the following quadratic form

$$Q(\mathbf{x}) = a(x_1^2 + bx_2^2 + x_3^2) + 2\sqrt{b}x_1(x_2 + x_3).$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Find all the principal submatrices of A .

Solutions: (2 marks) The matrix of coefficients is:

$$A = \begin{pmatrix} a & \sqrt{b} & \sqrt{b} \\ \sqrt{b} & ab & 0 \\ \sqrt{b} & 0 & a \end{pmatrix}$$

(2 marks) Its principal submatrices are:

$$\begin{aligned} &3^{\text{rd}} \text{ Order: } \begin{pmatrix} a & \sqrt{b} & \sqrt{b} \\ \sqrt{b} & ab & 0 \\ \sqrt{b} & 0 & a \end{pmatrix} \\ &2^{\text{nd}} \text{ Order: } \begin{pmatrix} a & \sqrt{b} \\ \sqrt{b} & ab \end{pmatrix}, \begin{pmatrix} a & \sqrt{b} \\ \sqrt{b} & a \end{pmatrix} \text{ and } \begin{pmatrix} ab & 0 \\ 0 & a \end{pmatrix}, \\ &1^{\text{st}} \text{ Order: } (a), (ab) \text{ and } (a). \end{aligned}$$

- (b) (6 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *negative semidefinite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is negative semidefinite?

Solutions: (3 marks) $Q(\mathbf{x})$ is negative semidefinite if $Q(\mathbf{x}) \leq 0$ for any $\mathbf{x} \in \mathbb{R}^3$.

(3 marks) To, check if $Q(\mathbf{x})$ is negative we need to verify that all odd principal minors are non-positive and that all even principal minors are non-negative.

- (c) (15 marks) Find the values of the parameter values, a and b , such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*? Then, plot these regions in $\mathbb{R} \times \mathbb{R}_+$.

Solutions: (5 Marks) $Q(\mathbf{x})$ is positive semidefinite if all principal minors are positive which requires:

$$a \geq 0, \quad ab \geq 0, \quad a^2b - b \geq 0, \quad a^2 - b \geq 0, \quad a^2b \geq 0, \quad a^3b - ab - ab^2 \geq 0.$$

For these to hold at once it must be that either $a = b = 0$ or

$$a \geq 1, \quad b \geq 0, \quad b \leq a^2 - 1.$$

(5 Marks) $Q(\mathbf{x})$ is negative semidefinite if all principal minors suitably alternate in sign which requires:

$$a \leq 0, \quad ab \leq 0, \quad a^2b - b \geq 0, \quad a^2 - b \geq 0, \quad a^2b \geq 0, \quad a^3b - ab - ab^2 \leq 0.$$

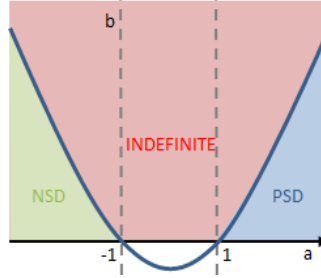
For these to hold at once it must be that either $a = b = 0$ or

$$a \leq -1, \quad b \geq 0, \quad b \leq a^2 - 1.$$

(2 Marks) Consequently, $Q(\mathbf{x})$ is indefinite whenever it is not semidefinite. Thus, $Q(\mathbf{x})$ is indefinite whenever

$$b > a^2 - 1 \text{ for } (a, b) \in \mathbb{R} \times \mathbb{R}_+ \setminus (0, 0).$$

(3 Marks) The plot below depicts the answer



5. (30 marks) Let $f : U \rightarrow \mathbb{R}$ be a function defined on a convex subset of $U \subset \mathbb{R}$.

- (a) (10 marks) State a definition of *concavity* for an arbitrary function f . Then, state a definition of *concavity for a twice differentiable function* f , which is based on the Hessian of f .

Solutions: (5 marks) The function f is concave if for any $x, y \in \mathbb{R}$

$$f(ty + (1 - t)x) \geq tf(y) + (1 - t)f(x) \text{ for all } t \in [0, 1].$$

(5 marks) The twice differentiable function f is concave if for any $x \in \mathbb{R}$

$$f''(x) \leq 0.$$

- (b) (20 marks) Prove that the two definitions you gave in part (b) are equivalent provided that f is twice differentiable.

Solutions: (5 marks) To show that the two definitions are equivalent first note that as f is differentiable, for any $x, y \in \mathbb{R}$ and any $t \in [0, 1]$,

$$\begin{aligned} tf(y) + (1 - t)f(x) \leq f(ty + (1 - t)x) &\Leftrightarrow t(f(y) - f(x)) + f(x) \leq f(x + t(y - x)) \Leftrightarrow \\ f(y) - f(x) \leq \frac{f(x + t(y - x)) - f(x)}{t} &\Leftrightarrow f(y) - f(x) \leq \frac{f(x + h) - f(x)}{h}(y - x), \end{aligned}$$

where $h = t(y - x)$. Taking limits when $h \rightarrow 0$, the latter implies that

$$f(y) - f(x) \leq f'(x)(y - x). \quad (1)$$

(5 marks) To prove the converse, let for any $x, y \in \mathbb{R}$ and any $t \in [0, 1]$, let $z = ty + (1 - t)x$. By applying equation (1), we have that

$$\begin{aligned} f(y) - f(z) &\leq f'(z)(y - z), \\ f(x) - f(z) &\leq f'(z)(x - z). \end{aligned}$$

Summing the first inequality multiplied by t to the second multiplied by $(1 - t)$, we find the result

$$[f(y) - f(z)]t + [f(x) - f(z)](1 - t) \leq 0 \Leftrightarrow tf(y) + (1 - t)f(x) \leq f(z).$$

(5 marks) To prove that equation (1) implies $f''(x) \leq 0$, take any $x, y \in \mathbb{R}$ and note that by (1)

$$f(y) - f(x) \leq f'(x)(y - x) \quad \text{and} \quad f(x) - f(y) \leq f'(y)(x - y).$$

Summing these and dividing by $-(y - x)^2$ implies that

$$\frac{f'(y) - f'(x)}{y - x} \leq 0.$$

The limit of the last expression when $y \rightarrow x$ yields $f''(x) \leq 0$.

(5 marks) To prove the converse, pick $y \geq x$, and note that by Fundamental Theorem of Calculus and $f'' \leq 0$, we have that

$$f(y) - f(x) = \int_x^y f'(t) dt \leq \int_x^y f'(x) dt = f'(x)(y - x).$$

6. (45 marks) For $a > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & y - e^x \\ \text{s.t.} \quad & ay \leq a^2x + 1 \\ & y \geq 0, x \geq 0. \end{aligned}$$

- (a) (5 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither. Can you solve this optimization problem using the Kuhn-Tucker method? That is, for what values of $a > 0$ is the Slater condition satisfied?

Solutions: (1 marks) The constraints are obviously linear and thus both convex and concave. (2 marks) The objective function is concave, as its Hessian is negative semidefinite,

$$D^2f(x, y) = \begin{pmatrix} 0 & 0 \\ 0 & -e^x \end{pmatrix}, \quad |D^2f(x, y)| = 0 \geq 0, \quad -e^x < 0 \text{ and } 0 \leq 0.$$

(2 marks) When the constraint set has a non-empty interior, we can adopt the Kuhn-Tucker approach to solve the problem. This is the case for all $a > 0$, since the point $\tilde{x} = \varepsilon$ and $\tilde{y} = a\varepsilon$ is always in the interior of the constraint set, as

$$a\tilde{y} = a^2\varepsilon < a^2\varepsilon + 1 = a^2\tilde{x} + 1, \quad \tilde{x} > 0, \quad \tilde{y} > 0.$$

- (b) (10 marks) Suppose the Slater condition is satisfied, write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian?

Solutions: (7 marks) The necessary first order conditions for the solution of the constrained maximization problem are:

$$\begin{aligned} 0 &= 1 - a\lambda_1 + \lambda_2, \\ 0 &= -e^x + a^2\lambda_1 + \lambda_3, \\ 0 &= \lambda_1(ay - a^2x - 1) = \lambda_2y = \lambda_3x, \\ 0 &\geq ay - a^2x - 1, \quad 0 \geq -y, \quad 0 \geq -x, \\ 0 &\leq \lambda_1, \quad 0 \leq \lambda_2, \quad 0 \leq \lambda_3. \end{aligned}$$

(3 marks) Any solutions to these conditions is maximizer of the Lagrangian, because the Lagrangian is a sum of concave functions by the arguments in (a) and because Slater condition holds.

(c) (20 marks) Solve the constrained optimization problem in terms of a .

Solutions: Consider the various supports:

- (3 marks) At a maximum it must be that $\lambda_2(a) = 0$. If $\lambda_2 > 0$ and $y = 0$, the objective could be improved while fulfilling the constraints for any $x \geq 0$, since

$$f(x, 0) = -e^x < ax + 1/a - e^x = f(x, ax + 1/a).$$

- (3 marks) At a maximum, it must be that $\lambda_1 > 0$, because $\lambda_1(a) = 1/a$ by the first KKT condition coupled with $\lambda_2(a) = 0$.
- (7 marks) Next consider the case in which $\lambda_1, \lambda_3 > 0$, while $\lambda_2(a) = 0$. If so, the two binding constraints would imply that

$$y(a) = 1/a \quad \text{and} \quad x(a) = 0.$$

Solving the two first order conditions for goods x and y would imply that

$$\lambda_1(a) = 1/a \geq 0, \quad \lambda_3(a) = 1 - a \geq 0.$$

But then $\lambda_3(a) \geq 0$ would require $a \leq 1$. Thus, there is a solution on this support if and only if $a \leq 1$.

- (7 marks) Finally, consider the case in which $\lambda_1 > 0$ while $\lambda_2(a), \lambda_3(a) = 0$. If so, solving the two first order conditions for goods x and y would imply that

$$\lambda_1(a) = 1/a \geq 0, \quad x(a) = \log(a) \geq 0,$$

which in turn requires $a \geq 1$ for $x(a) \geq 0$. But then, the binding constraint would imply that

$$y(a) = a \log(a) + 1/a \geq 1/a > 0.$$

(d) (10 marks) State the Envelope Theorem and use it to find the slope of value function with respect to the parameter a . Is the value function concave or convex? For what value of $a > 0$ is the value function minimized?

Solutions: (3 marks) The Envelope Theorem states that if $(x(a), y(a), \lambda(a))$ are differentiable functions and the constraint qualification holds, then

$$\frac{dv(a)}{da} = \frac{\partial L(x(a), y(a), \lambda(a))}{\partial a} = \lambda_1(a)(2ax(a) - y(a))$$

(2 marks) If $a < 1$, $x(a) = 0$ while $y(a) = \lambda_1(a) = 1/a$. Thus, the value decreases in a

$$\frac{dv(a)}{da} = \lambda_1(a)(2ax(a) - y(a)) = -1/a^2 < 0.$$

(2 marks) If $a > 1$, $x(a) = \log(a)$, $y(a) = a \log(a) + 1/a$ while $\lambda_1(a) = 1/a$. Thus, the value first decreases and then increases in a

$$\begin{aligned} \frac{dv(a)}{da} &= \lambda_1(a)(2ax(a) - y(a)) \\ &= \log(a) - 1/a^2. \end{aligned}$$

(3 marks). So the only critical point of the value function is the level $a_* > 1$ at which

$$a_*^2 \log(a_*) = 1.$$

Moreover, this point is indeed a minimum as the value function is convex at a_* since

$$\frac{d^2v(a_*)}{da^2} = \frac{a_*^2 + 2}{a_*^3} > 0.$$